CSA06507-Generative AI

Lab Experiment's

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Unit 2: Lab Programs for Prompting

Experiment 6: Text Generation Using OpenAI LLM API

1. Aim

To develop a Python program in Google Colab that connects to a Large Language Model (LLM) using the OpenAI API and generates text based on a user-provided prompt. The program accepts input from the user and displays the AI-generated response.



OpenAI API

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2. Algorithm

1. Start the program.
2. Install the OpenAI Python library.
3. Import the required library and configure the OpenAI API key.
4. Create an OpenAI client.
5. Get a text prompt from the user.
6. Send the prompt to the LLM using the API.
7. Receive the generated response.
8. Display the generated text.
9. Stop the program.

3. Python Code (

Experiment 6: Text Generation Using OpenAI LLM API

Step 1: Install OpenAI library

!pip -q install openai

Step 2: Import libraries

from openai import OpenAI

from getpass import getpass*# Step 3: Enter API key securely*

api_key = getpass("Enter your OpenAI API key: ")

```
# Step 4: Create OpenAI client
client = OpenAI(api_key=api_key)
# Step 5: Get prompt from user
prompt = input("Enter your prompt: ")
# Step 6: Generate text using the LLM
try:
    response = client.responses.create(
        model="gpt-5.5",
        input=prompt
    )

    # Step 7: Display generated text
    print("\nGenerated Text:")
    print(response.output_text)

except Exception as e:
    print("Error:", e)
    print("Check your API key, model access, and API usage.")
```

Important: Get your API key from the OpenAI platform. Never share it publicly or paste it into a notebook you share with others. API usage may incur charges.

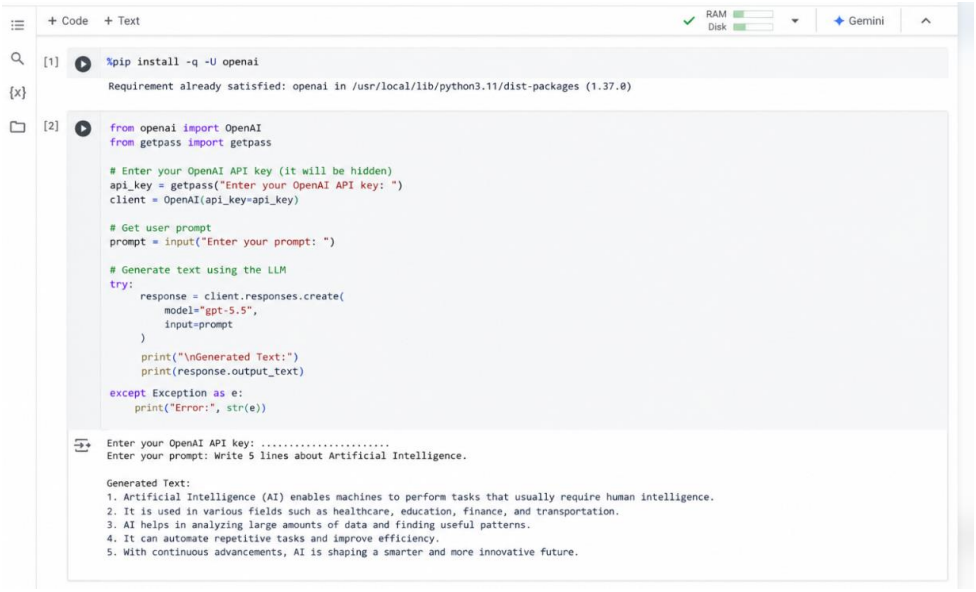
4. Input

Example user input:

Plain text

Enter your prompt: Write a short paragraph about Artificial Intelligence.

5. Output



6. Result

The Python program connects to an OpenAI Large Language Model through the API, accepts a user-provided prompt, and displays the generated text. Thus, text generation using an LLM API is demonstrated successfully when a valid API key and model access are available.